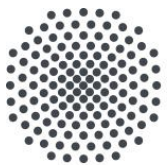


Low-Code Quantum Application Development: From Visual Modeling to Real Quantum Execution



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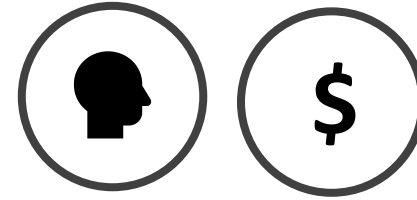
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Institute for Architecture of Application Systems

Fundamentals

- Classical Bit

- like a coin, either 0 (head) or 1 (tail)



- Qubit

- can be both tails and heads at the same time



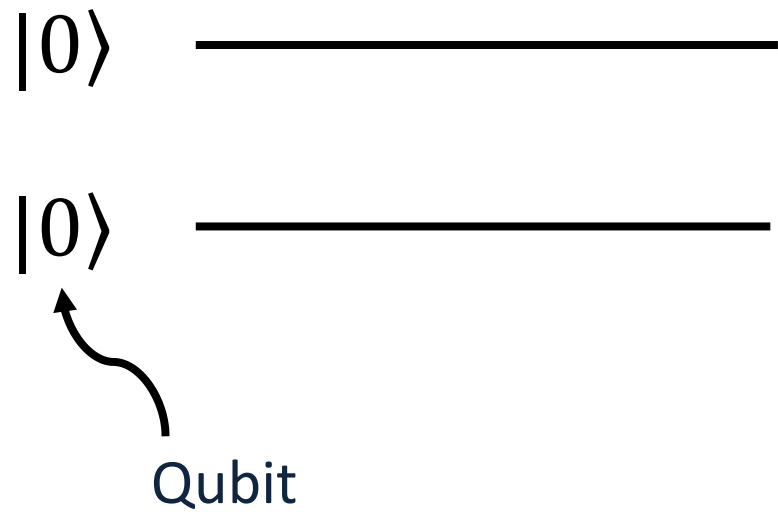
- Superposition

- many states at the same time

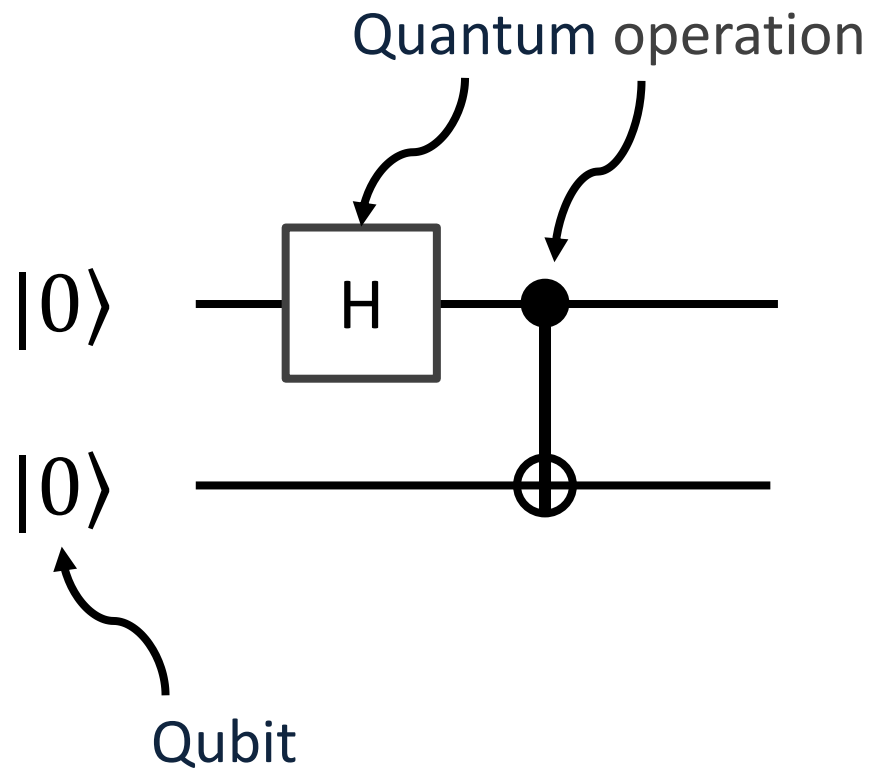
- Measurement

- forces the qubits to collapse into a definite classical value

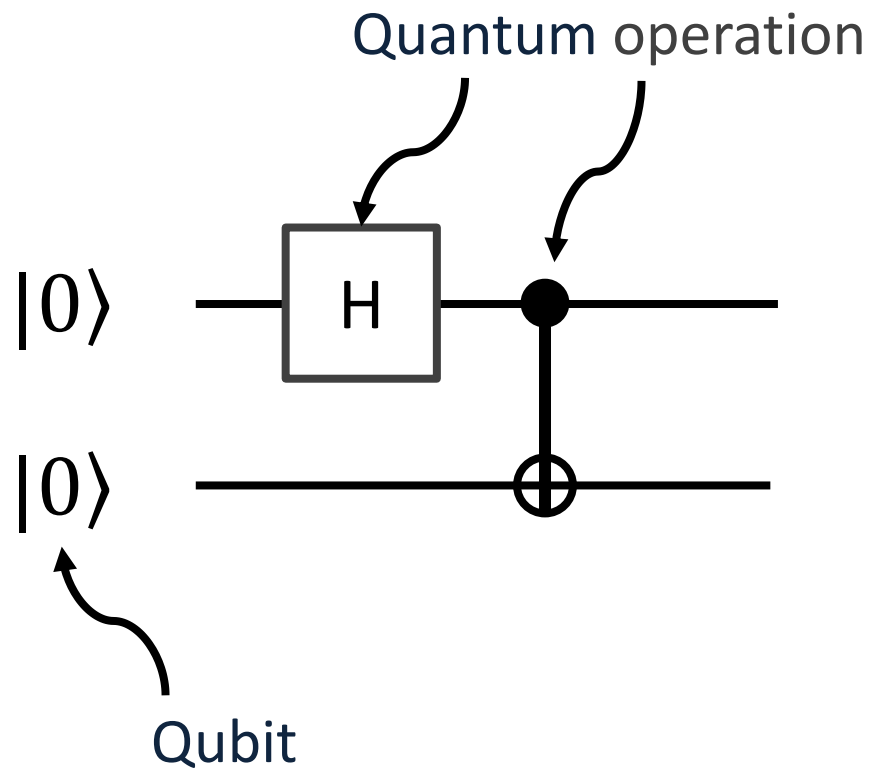
Fundamentals



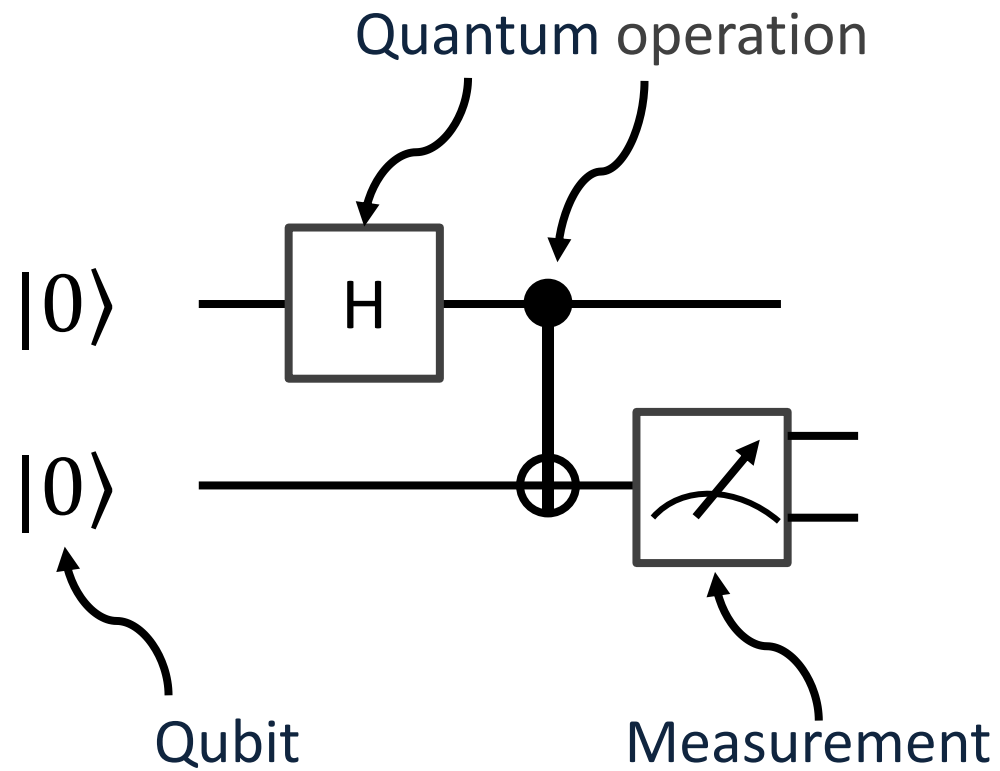
Fundamentals



Fundamentals



Fundamentals



Fundamentals

H

X

Z

Y

S

T

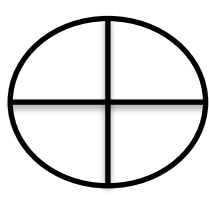
$T^\dagger$

$S^\dagger$

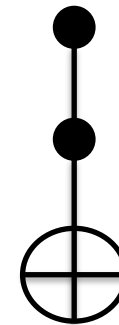
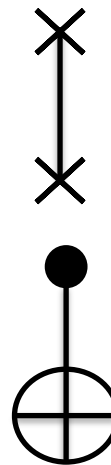
$\sqrt{X}$

$\sqrt{X}^\dagger$

P

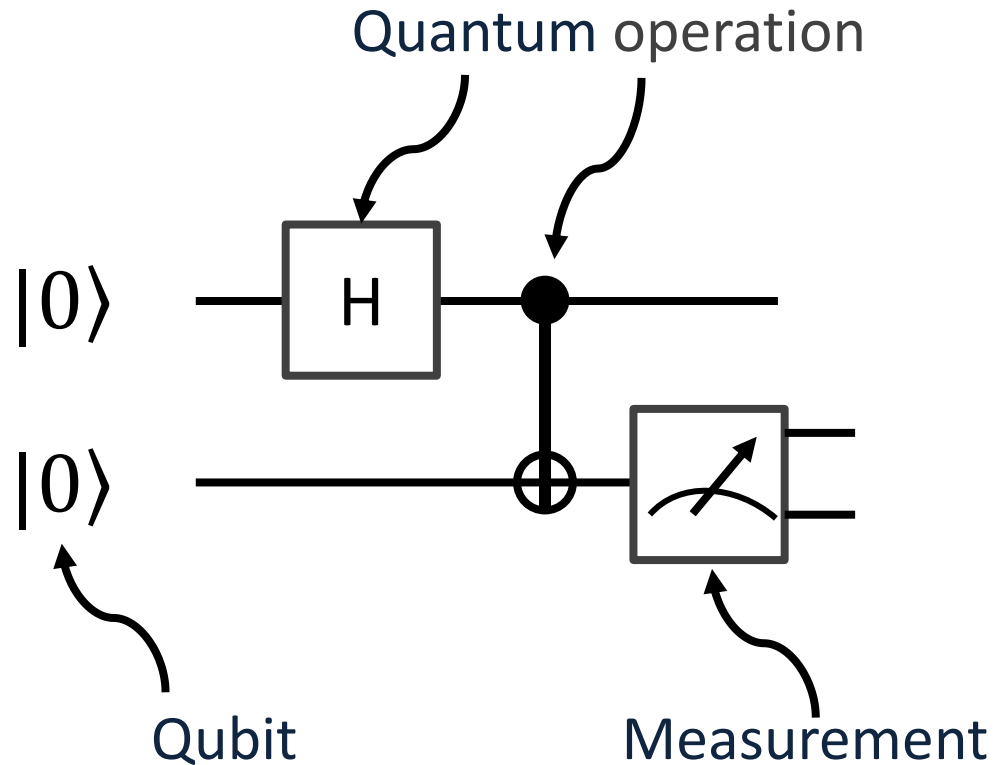


SWAP



...

Fundamentals



```
// Imports
```

```
qc = QuantumCircuit(2, 1)
```

```
qc.h(0)
```

```
qc.cx(0, 1)
```

```
qc.measure(0, 0)
```

Fundamentals

```
// Imports
```

```
qc = QuantumCircuit(2, 1)
```

```
qc.h(0)  
qc.cx(0, 1)
```

```
qc.measure(0, 0)
```

Qiskit

```
OPENQASM 3.0;
```

```
include "stdgates.inc";
```

```
qubit[2] q;  
bit[1] c;
```

```
h q[0];  
cx q[0], q[1];
```

```
c[0] = measure q[0];
```

OpenQASM

```
// Imports
```

```
circuit = Circuit()
```

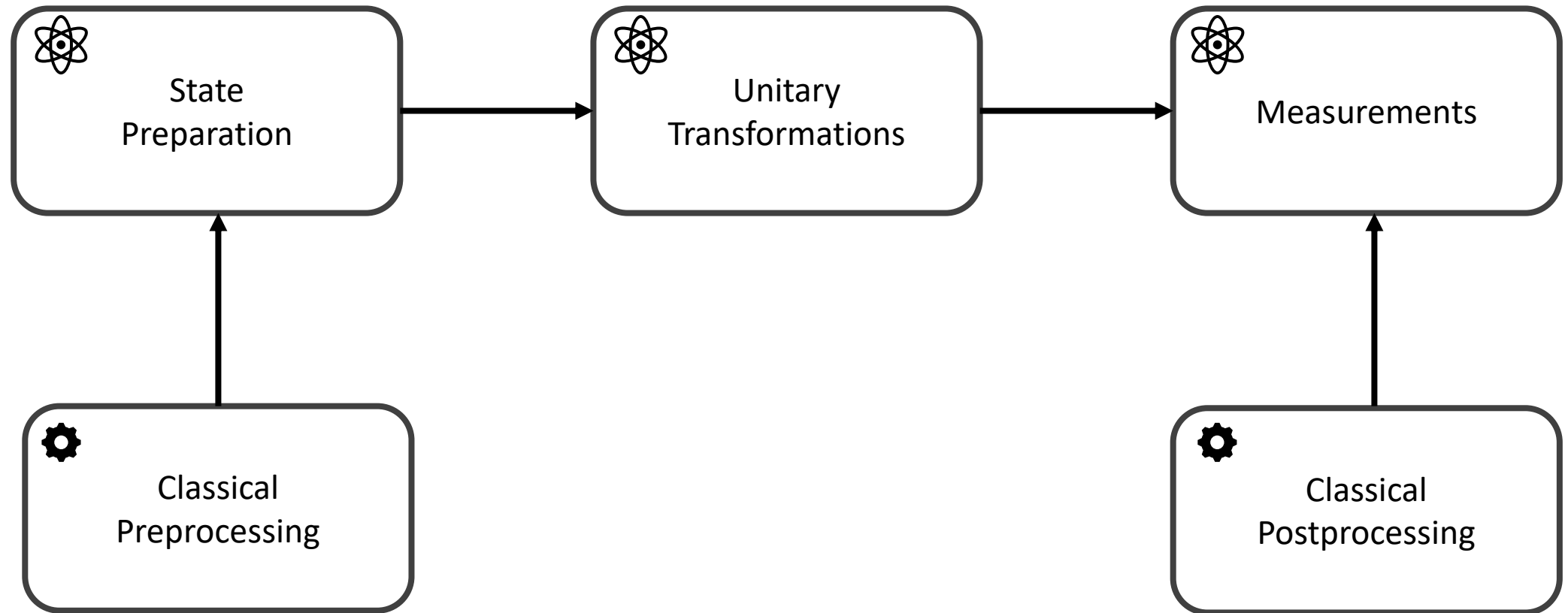
```
circuit.h(0)  
circuit.cnot(0, 1)
```

```
circuit.measure(0)
```

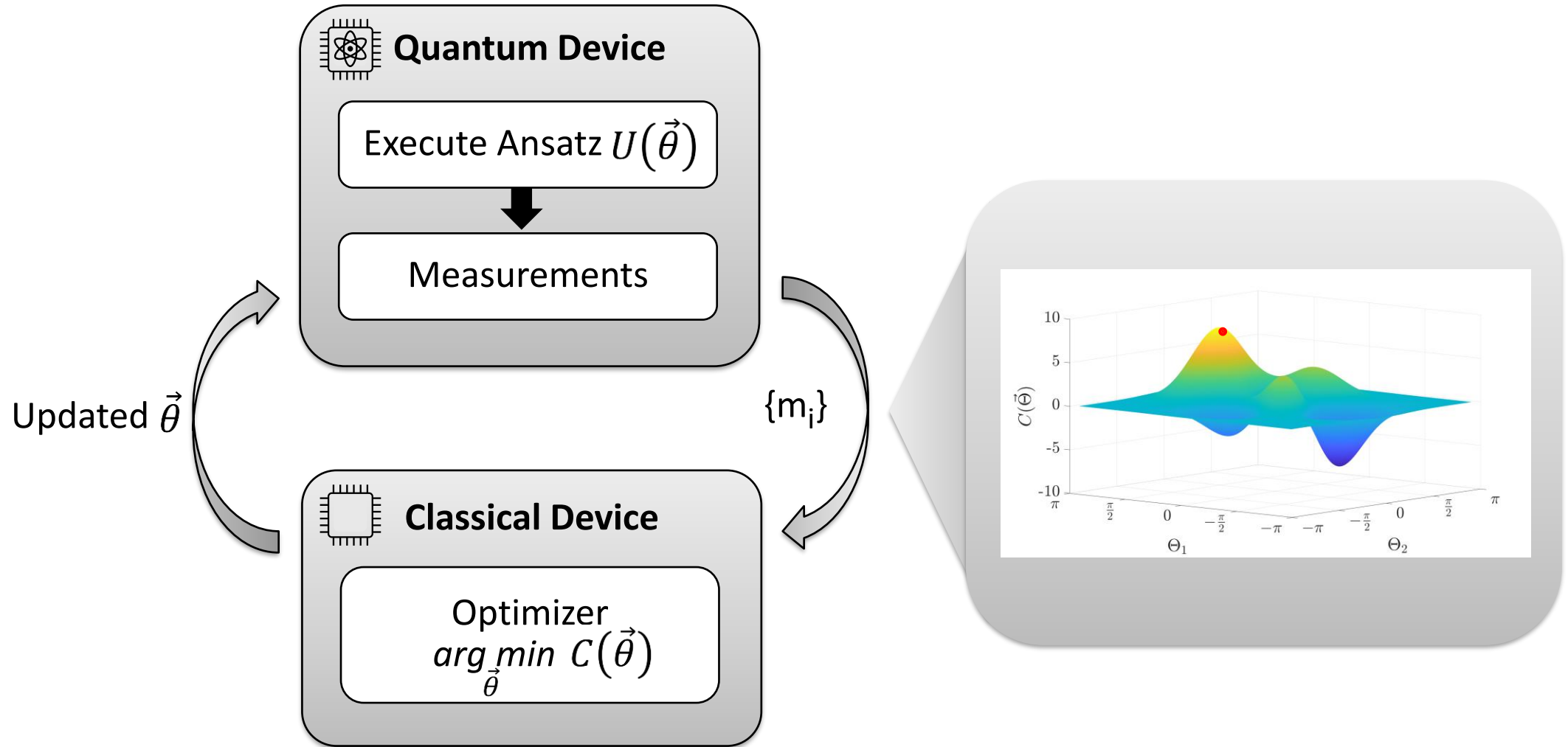
Braket

Fundamentals

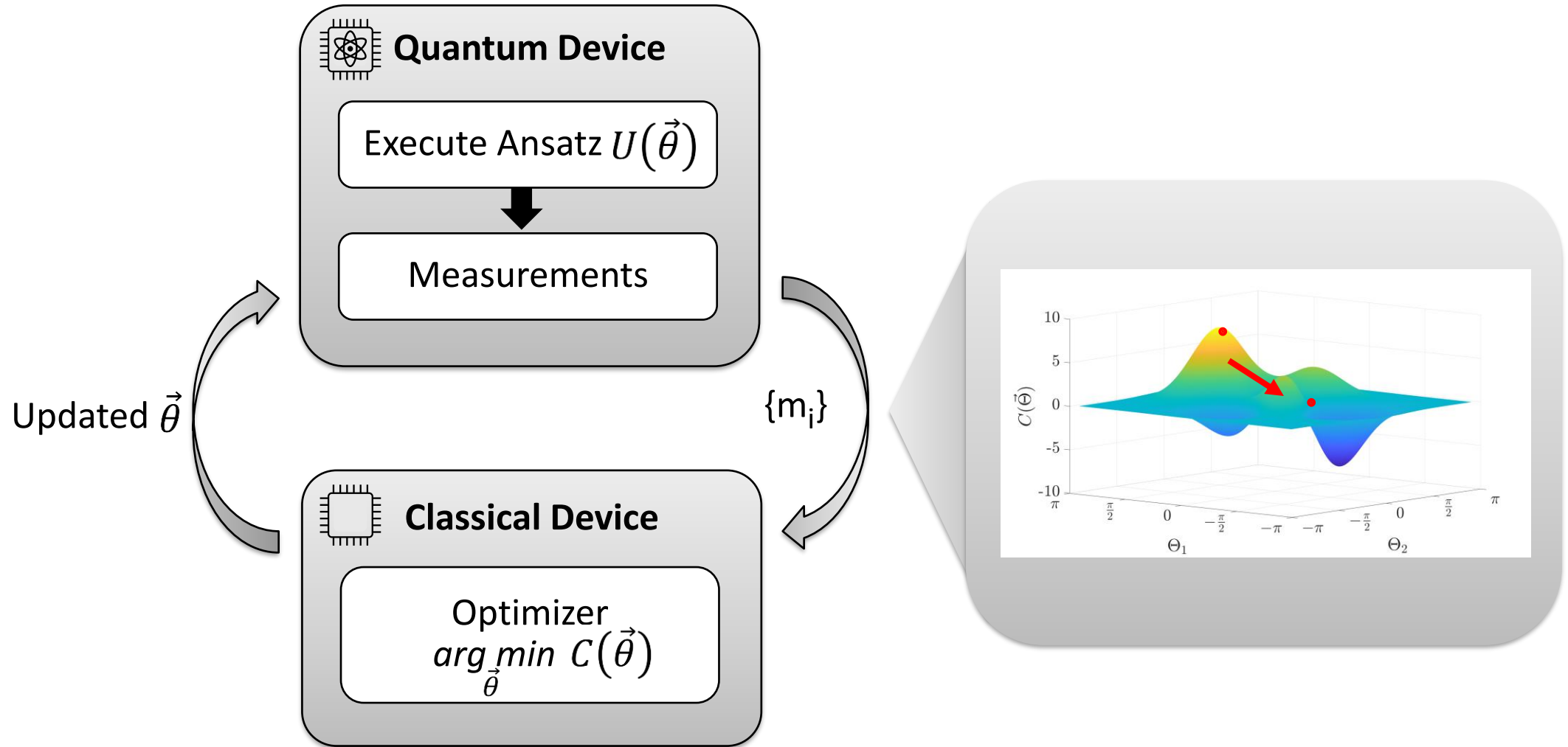
- But quantum circuits are not enough!



Motivation



Motivation



Thank you for your attention!